VARDHAMAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Affiliated to JNTUH, Approved by AICTE, Accredited by NAAC with A++ Grade, ISO 9001:2015 Certified

Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India.

ASSIGNMENT -1

Branch & Section:	III B.Tech. I Semester (CSD) R22	Academic Year:	2026-27
Course:	OTML A8751	Total Marks:	10 M
Name of the Student:		Roll No:	
Date of submission:		Signature of the faculty	
Name of the CourseInstructor:			

Instructions:

A. Answer all the questions, and each question carries 5 marks.
B. Refer to the textbooks/reference books listed below.
C. You must print this page and start answering only in A4 sheets including this page.
D. Note:- Last date for submission is 25.07.2026

Q. No.	Questions	CO#	BL#
1	State Gradient Descent Function, derive the iteration for $\theta^{(1)},\theta^{(2)},\theta^{(3)}$ by using X = {2,4,6} and Y = {2,8,10} where learning value $\theta = 0.2$ and $\theta_0 = 0$	CO1	L1
2	How can we verify if an optimization problem is convex? Discuss the role of the Hessian and conditions for convexity.	CO1	L1
3	Discuss the concept of Empirical Risk Minimization (ERM) and elaborate on its relationship with loss functions	CO1	L1
4	Fit a straight-line Y = θ X to the data points X = {1,2} and Y = {2,3} using the vectorized general solution. Clearly construct the feature matrix for the observations	CO1	L1
5	State and derive the normal equation for linear regression: $\theta = (X^TX)^{-1} X^T y$	CO1	L2
6	Given a prior distribution $p(\theta) = N(\mu_0, \Sigma_0)$ and a likelihood function $p(y\|X,\theta) = N(X\theta, \sigma^2 I)$, derive the mathematical expressions for the posterior mean μ_post and the posterior covariance Σ_post	CO1	L2

START WRITING ANSWERS FROM HERE